

Proposal of a Multimodal and Multiscale Framework for Sleep Apnea Diagnosis

Multiscale and multimodal biocybernetics – Academic year 2024/2025

Federico Bordin
337573

Enrico Conti
330765

Lorenzo Fracchia
314747

Marcelo Conrado Venturini
338023

Abstract - Sleep Apnea (SA) is a common sleep disorder characterized by repeated episodes of partial or complete breathing cessation during sleep, which can lead to significant health risks such as cardiovascular complications and chronic fatigue. We introduce a cost-effective, home-based, multimodal and multiscale diagnostic framework that leverages different key physiological signals to detect apnea events. Our system uses a Gated Recurrent Unit (GRU) network with a target replication strategy to capture temporal dependencies in the data. The model, to be trained on a newly curated dataset annotated by sleep specialists, addresses class imbalance with macro F1-score evaluation and offers explainability by highlighting significant signal segments. This framework promises a more accessible and comfortable alternative for accurate SA diagnosis.

Index Terms - Sleep Apnea Diagnosis, Gated Recurrent Unit (GRU), Multimodal and Multiscale Framework, PAT, SPO2, snoring.

I. INTRODUCTION

A. Introduction of the pathology

Obstructive Sleep Apnea (OSA), is a disorder characterized by repeated episodes of complete or partial cessation of airflow through the upper airways, caused by their recurrent collapse during sleep [1]. This results in irregular breathing, observed apneas, shortness of breath with frequent awakenings, restless sleep, snoring, headaches, obesity (common among adults), impaired cognitive function, daytime sleepiness, reduced quality of life, anxiety and depression. However, most of these symptoms aren't sufficient for diagnosis, as the clinical presentation does not correlate well with standardized sleep study results. Additionally, determining the severity of the condition is challenging.

OSA is considered a chronic condition, as it meets the criteria established by the *World Health Organization* for such diseases: it is long-lasting, tends to progress slowly, and is not transmissible.

OSA can occur at any age, with a higher prevalence between 40 and 60 years. In adults, it is estimated that the percentage of people suffering from this disease ranges from 26.2–52.6% affecting mostly men [1]. The severity of this disease is evaluated using the apnea-hypopnea index (AHI) that is the sum of this events in the laps of an hour of sleep and can be classified as mild apnea ($5 < \text{AHI} < 15$), moderate apnea ($15 < \text{AHI} < 30$) and severe apnea ($\text{AHI} > 30$) [2].

B. Symptoms and their impact on health

Recent studies indicate that OSA is also associated with other chronic conditions, such as cardiac arrhythmias other than atrial fibrillation, cognitive and mood disorders, depressive syndrome, insomnia, bronchial asthma, kidney failure, cancer, and liver steatosis [2].

According to the report "Global Surveillance, Prevention and Control of Chronic Respiratory Diseases: A Comprehensive Approach" published in 2007, over 100 million people worldwide are affected by OSA, although more than 75% of cases remain undiagnosed. Which is why the importance of a simple, low cost and effective tool for diagnosing this disease is essential.

In contrast to obstructive sleep apnea (OSA), in which ongoing respiratory efforts are observed, central sleep apnea (CSA) is defined by a lack of respiratory effort during cessations of airflow [21].

II. DATA ACQUISITION AND REPRESENTATION

A. Overview

Polysomnography (PSG) is the current gold standard test for OSA, providing the AHI as a function of disease severity using a variety of signals. PSG present a high sensitivity (80%) and specificity (97%) [22], but has on the other side limited access, especially in rural and economically disadvantaged areas. It is expensive, time consuming and necessitates of a specific laboratory in a clinical setting or an hospital.

This framework aims at providing a more affordable and comfortable home-based manner of diagnosing for Sleep Apnea: for these reasons only the Oxygen Saturation (SPO2), Peripheral Arterial Tomography (PAT) and snoring signals will be considered (Figure 1, panel a). While the information contained in these signals can be enough to identify apnea-hypopnea events, there are some other signals that are recommended and are useful like electrocardiogram (ECG), electroencephalogram (EEG) [6] or even imaging modalities like MRI [14]. However, having in mind the objective stated above, we decided not to include them since the information they could provide did not outweigh the costs and complications and since OSA is an underdiagnose condition, we believe simplicity is of the utmost importance.

B. SPO2 signal

Apnea events involving the cessation of breathing cause oxygen desaturation. The severity of oxygen desaturation depends on the duration of the event and the baseline oxygen saturation level [7]. The exact drop in oxygen saturation is determined by the oxygen dissociation curve and the placement of the sensor [8]. This can lead to non-negligible variability in the signal, but the signal itself is very intuitive and therefore highly valued as a measure for sleep apnea [4]. According to the *American Academy of Sleep Medicine* manual, the drop in oxygen saturation is a diagnostic criterion for defining hypopnea events [6]. As is known today, the count of apnea and hypopnea events (i.e., AHI) is not the only measure to determine the severity of

sleep apnea [9]; oxygen saturation is a good candidate because hypoxia is known to impair physiological functions and cognition [10] and correlates with arterial hypertension in patients with sleep apnea [11].

Smartwatches equipped with LED technology on the underside of the watch can record oxygen saturation in real time, provided that good processing of the pulse wave signal is implemented [12].

C. Peripheral Arterial Tonography

The Peripheral Arterial Tonometry signal (PAT) is obtained using plethysmographic methods. This technique facilitates the accurate recording of the pulsatile volume changes of the arteries of peripheral vascular beds at the distal end of the fingers over sustained periods of time and is mainly invariant with age [3,4]. This signal measures one of the most important sympathetic-related parameters which is digital vasoconstriction [3]. Measuring the sympathetic nervous system activity is crucial because it is elicited by sleep arousal which is a hallmark of sleep disordered breathing [3,5]

Airflow obstruction in sleeping patients with OSA leads to an increased digital vasoconstriction and the magnitude of the PAT signal attenuation depends on the degree of the airflow obstruction. This attenuation can be noticed even in the absence of a detectable EEG arousal [5].

It has been proven that all the PAT attenuation events during the different phases of sleep are associated with activity in the alpha frequency range that coincides with a vasoconstrictory phase [3]. This means that we can obtain information about the alpha frequency without utilizing an EEG setup which is complicated and requires specific equipment [6].

The PAT amplitude attenuations have also been shown to correlate with the surges in blood pressure at the termination of apneic events and with the increased heart rate [3,6]

D. Snoring

It is commonly stated that snoring accompanies obstructive sleep apnea (OSA). However, regular snoring is bothersome but not characteristic of sleep apnea. It may reflect vibrations of the upper airway tissues and not necessarily an obstruction of the upper airway. When upper airway obstruction occurs, snoring becomes irregular, and the sound changes. Analyzing the details of snoring sounds can therefore support the diagnosis of sleep apnea [1].

In this case, a microphone is placed above the trachea, and the recorded signal is referred to as a tracheal sound. A sophisticated analysis of the tracheal sound allows for extracting information about respiratory flow, including upper airway obstruction with characterization of obstructive apnea events and an estimation of respiratory effort. This type of sensor has been integrated into a French system used for PSG in the diagnosis of sleep-disordered breathing [13]. Recent studies have shown that automated tracheal sound analysis can effectively identify sleep apnea-hypopnea syndrome [14].

E. Multimodal and Multiscale Approach

In our framework multimodality is granted by the fact that each sensor modality targets a different physiological

facet associated with sleep apnea. By fusing these varied data streams, our method compensates for the limitations of any single modality and delivers a richer diagnostic picture.

Furthermore, the signals integrated into our system operate at different time scales and frequency resolutions, enabling the capture of both rapid and gradual physiological changes. By incorporating measurements that reflect both systemic (oxygen saturation) and localized (vasoconstriction via PAT) responses, our framework addresses the multi-scale nature of sleep apnea pathogenesis. This layered approach allows for the identification of apnea events through macro-level indicators (snoring) as well as micro-level cues, offering a more detailed understanding of disease severity and progression.

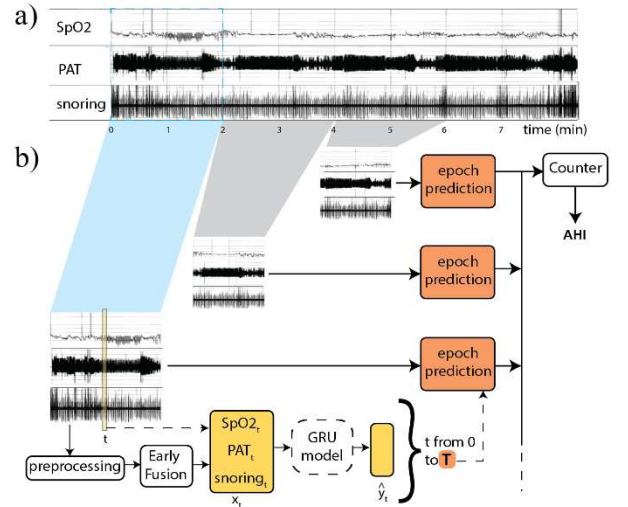


Figure 1. Panel a: SpO2, PAT and snoring raw signals. Panel b: Pipeline of the epoch segmentation, epoch prediction through the GRU model and consequent derivation of the AHI.

III. DATASET

A. Dataset Creation and Data Collection

We believe that a dataset specifically tailored for our method would greatly enhance the results. For this reason, in our framework, we propose the creation of a novel dataset for the development of an automatic diagnostic method for sleep apnea. The dataset will be derived from the raw physiological signals obtained through a portable device at home, specifically *SpO2*, *peripheral arterial tonometry (PAT)*, and *snoring recordings* during at-home sleep tests. The proposed dataset will be annotated by sleep specialists and will include a diverse range of patients. Ethical considerations will be addressed through proper consent and approval procedures.

The aid of the specialists will be fundamental in the data validation and preprocessing too.

1) Data Collection Protocol

The data will be divided into time epochs of 2 minutes. This will allow us to count the number of sleep apnea events per hour, an important indicator of the severity of the disorder. Each epoch will be annotated by two experienced sleep specialists to classify the data as *normal breathing patterns*, *obstructive sleep apnea* or *central sleep*

apnea. In case of opposing opinions, a third specialist will be the final decider.

We aim to achieve a dataset that includes both patients diagnosed with sleep apnea and healthy controls, following the natural distribution of patients seeking diagnostic tests.

B. Dataset preparation

1) Data validation and cleaning

The specialists will also be tasked with the reporting of any problem in the data acquired by the rest of the patients, which may have been caused by an erroneous use of the device, such as an incorrect placement or battery problems. This report will be in the form of sample interval marking.

Epochs that contain less than 10% of samples deemed as corrupted will be restored in preprocessing with a forward filling approach, while epochs with more than 10% of corrupted samples will be discarded.

In case of patients' data where more than 15% of epochs were discarded, the whole patient data will be marked as unreliable and therefore not used.

2) Data preprocessing

After the cleaning, as previously mentioned, some of the epochs will receive a forward filling manipulation to replace the corrupted sample segments. Data values will be rescaled in a 0 to 1 interval in accordance with acceptable clinical reference values, taking on account age and gender. Since the data will be acquired in a domestic context, it is imperative to apply a notch filter (50hz in EU) to the time series, as capacitive pairing with the electrical grid is very likely.

As the acquired signals have different sampling frequency, it is necessary to equalize the number of samples in each epoch. The chosen method is to interpolate or resample each signal so that every channel contains a number of samples equal to the average original sample number.

3) Data subdivision

The last step for the preparation of our dataset is the data subdivision in training set, validation set and test set, which will be achieved by means of a traditional 80%, 10% and 10% subdivision.

IV. PROPOSED FRAMEWORK

The objective of the framework is to classify each 2 min-long epoch with a deep learning model as either a normal breathing, OSA or CSA episode. Additionally, the AHI is obtained from the predictions of all the epochs of an overnight subject's recording, which is used as an estimation of the severity of the pathology (mild, moderate or severe). A pipeline of the framework is shown in Figure 1, panel b.

A. Model description

We employ a Gated Recurrent Unit (GRU) architecture (Figure 2, panel a), which has shown to be well-suited to capture temporal dependencies in sequential data. Similarly to other RNN models like LSTMs, GRU networks can mitigate the vanishing gradient problem by means of gating mechanisms, while offering shorter training time and comparable performances [15]. Each GRU layer consists of two gates, the update gate (z_t) and the reset gate (r_t), which control the information flow from the input layer x_t to the

hidden vector h_t through both non-recurrent and recurrent connections, which retain information from previous steps (h_{t-1}). The input vector of the first layer contains the samples of all signals at time step t , therefore consisting in an early fusion of the different sources, while the hidden vector of the first layer becomes the input vector of the possible following layer. The last GRU layer is followed by a fully connected layer and a SoftMax, generating the output vector $\hat{y}_t$.

A target replication approach is integrated in the model, similarly to the one proposed by *Lipton et al.* in the context of multilabel classification from multivariate time series obtained from biological signals using LSTM architectures [16]. With this approach, for any given example (that is, a 2 min-long epoch) the model is not trained solely based on the output $\hat{y}_T$ at the last time step T ; instead, its label is replicated for each time step t and represents the target y_t for the t -th output of the GRU. Consequently, the loss function accounts for the average cost associated to each time step as well as the final cost at the end of the sequence. In such way, the network can learn to predict the target even at early steps of the sequence, without relying on the transmission of information across multiple steps to influence the final output. Due to the multi-class nature of the framework, the cost at a given time step is computed using the *categorical cross entropy (cce)* [17], and the overall loss function for a given sequence is computed as follows [16]:

$$\alpha \cdot \frac{1}{T} \sum_{t=1}^T cce(\hat{y}_t, y_t) + (1 - \alpha) \cdot cce(\hat{y}_T, y_T) \quad (1)$$

where $\alpha \in [0,1]$ represents the importance of the intermediate steps. Finally, for the prediction, the training is performed for 30 epochs using the Adam optimizer [18]. Parameters like the number of layers of the network, the number of hidden units per layer, the learning rate and the mini-batch size will be tested at early stages to find the best compromise between performances and computational time.

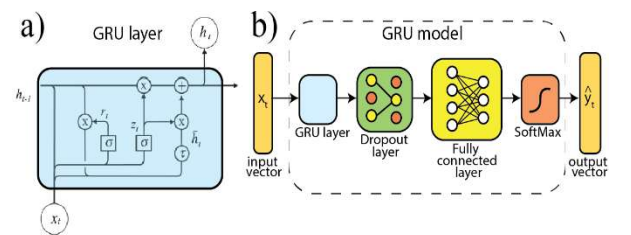


Figure 2. Panel a: schematic of a GRU layer (from *Erdenebayar et al.* [15]). Panel b: example of architecture with one GRU layer and dropout on the non-recurrent layers.

B. Regularization and optimization techniques

To avoid exploding gradients, we apply a ℓ_2 weight decay regularization, consisting in an additional penalty term in the loss function equal to the sum of squares of the weights, scaled by a weight decay factor λ . Additionally, to prevent overfitting, we employ drop out to the non-recurrent connections of the architecture [19]. Different dropout percentages, as well as different values of λ are explored during training and evaluated on the validation set

in the tuning process of the hyperparameters. Figure 2, panel b shows a schematic of the architecture.

C. Evaluation metrics

To account for the substantial imbalance of the dataset, in which the *normal breathing patterns class* is much more represented with respect to the two pathological classes of interest, the *macro F1-score* is chosen to assess the performances of the network throughout training, to ensure that all classes, regardless of their size, contribute equally to the evaluation [20]. Indeed, the *macro F1-score* is obtained by first computing the *F1-scores* independently for each class c from the *precision* (p) and *recall* (r) metrics and then averaging the scores across all classes. Specifically, the F1-score of the c -th class is computed as:

$$F1_c = 2 \cdot \frac{p_c \cdot r_c}{p_c + r_c} = 2 \cdot \frac{\frac{TP_c}{TP_c + FP_c} \cdot \frac{TP_c}{TP_c + FN_c}}{\frac{TP_c}{TP_c + FP_c} + \frac{TP_c}{TP_c + FN_c}} \quad (2)$$

Where TP_c , FP_c , and FN_c are respectively the number of true positives, false positives and false negatives of a given class c .

Since during the learning process the model might gradually lose its ability to generalise and it is not necessarily the most efficient at the last epoch, throughout the training the network is saved at each epoch, and the *macro F1-score* is computed on the validation set to find the best version of the system. Finally, the performances of the chosen model are evaluated on the test set, computing precision and recall for each class and the *macro F1-score*.

D. Explainability

The architecture produces a probability output at each instant and the model learns to classify also based on these intermediate steps. Therefore, after the training phase, when the model performs the prediction of a new sequence it is possible to visualize the evolution of the outputs through time as shown in Figure 3. This type of graph can give clearer insights into which segments of the example had a great contribution in the detection of a given class and highlight patterns that might be strongly correlated with the pathologies.

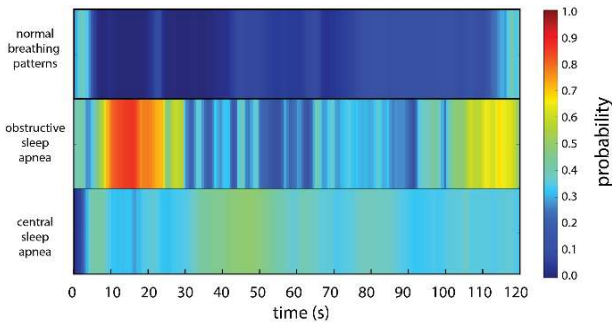


Figure 3. Graph of evolution through time of class probability outputs of a sequence whose final prediction is *obstructive sleep apnea* (image inspired by Lipton et al. [16]).

V. BIOCYBERNETIC LOOP

By combining the AI-generated prediction with the clinician's experience, it is possible to introduce a feedback mechanism based on a biocybernetic loop. The predictions are presented to a specialist via an intuitive interface, who

can either prescribe a PSG to confirm the diagnosis or directly recommend a treatment, depending on the severity of the symptoms.

To increase patient empowerment, it's possible to integrate the biocybernetic loop into a mobile application that serves as a central hub for data collection, processing and reporting, and allows the patient to access the reports.

Thanks to the continuous feedback of the biocybernetic loop, it's possible to facilitate and speed up personalised interventions. This approach bridges the gap between AI-driven diagnostics and clinical expertise, providing an innovative and efficient solution for OSA management.

VI. REFERENCES

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